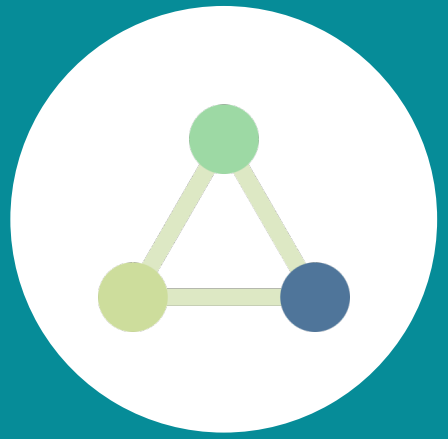
RESILIENT

Resilient Energy System Infrastructure Layouts for Industry, E-Fuels and Network Transitions



Dr Lukas Franken, TU Berlin, Germany

CETPartnership Annual Conference · Innovations in Action · 20 October 2026

Co-funded by the European Union

Introducing our Project

Resilient
Energy
System
Infrastructure
Layouts for
Industry,
E-Fuels and
Network
Transitions

FUNDING
CETPartnership 2022 call

COORDINATOR
TU Berlin, ENSYS

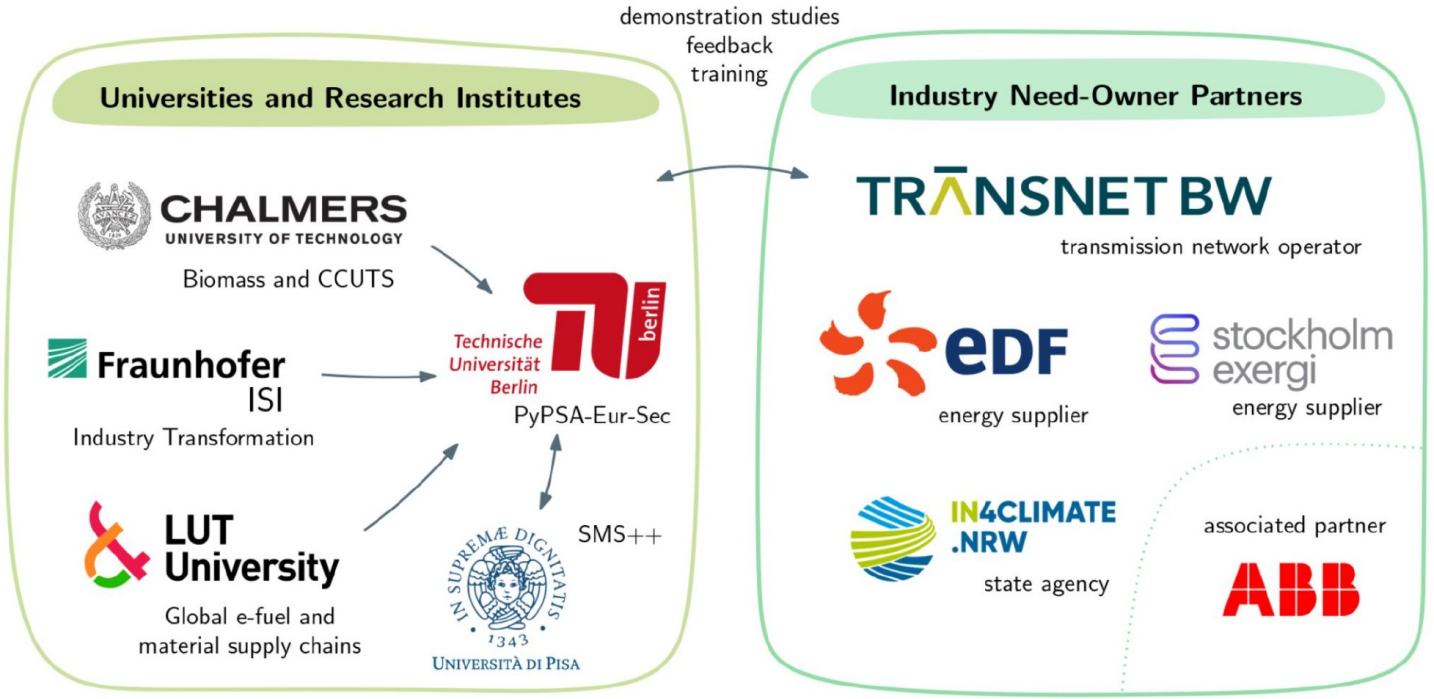
KICK-OFF
February 2024, Berlin

FINAL WORKSHOP
20 November 2026, Berlin

CONSORTIUM
10 partners, 5 countries

Co-funded by
the European Union

Our Consortium



10 partners | Germany · 4 | Sweden · 2 | Finland · 2 | Italy · 1 | France · 1

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Our Research Focus

UNCERTAINTIES

- Fuel & technology costs
- Hydrogen availability
- Grid & pipeline delays
- Industry restructuring
- E-fuel & material imports
- Renewables build-out
- Social acceptance

PyPSA-Eur

+ stochastic optimisation & CVaR
+ industry transformation
+ e-fuels, biomass & carbon capture
+ SMS++ decomposition

OUTPUT

Resilient infrastructure layouts

Case studies

- France
- Germany
- Sweden
- Finland

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Key Innovations

Open-source planning of energy infrastructure under uncertainty

17	24	9	~220
research papers	scientific talks	workshops	people trained

WHO BENEFITS

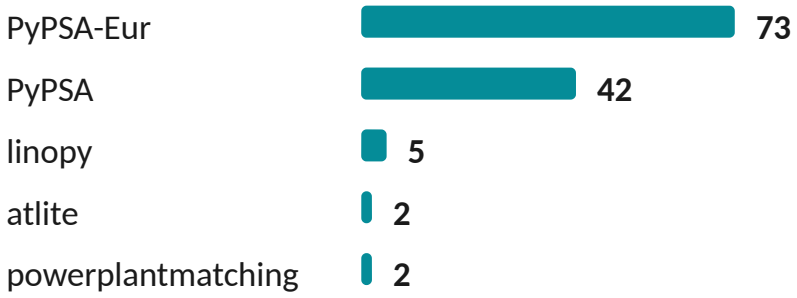
- Transmission system operators
- Energy suppliers
- Industry
- State agencies & policy
- Open-source modelling community

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Key Innovations

OPEN-SOURCE CODE

124 pull requests merged into the PyPSA ecosystem



Stochastic optimisation & CVaR
in PyPSA v1.0

Model dualisation
linopy v0.8

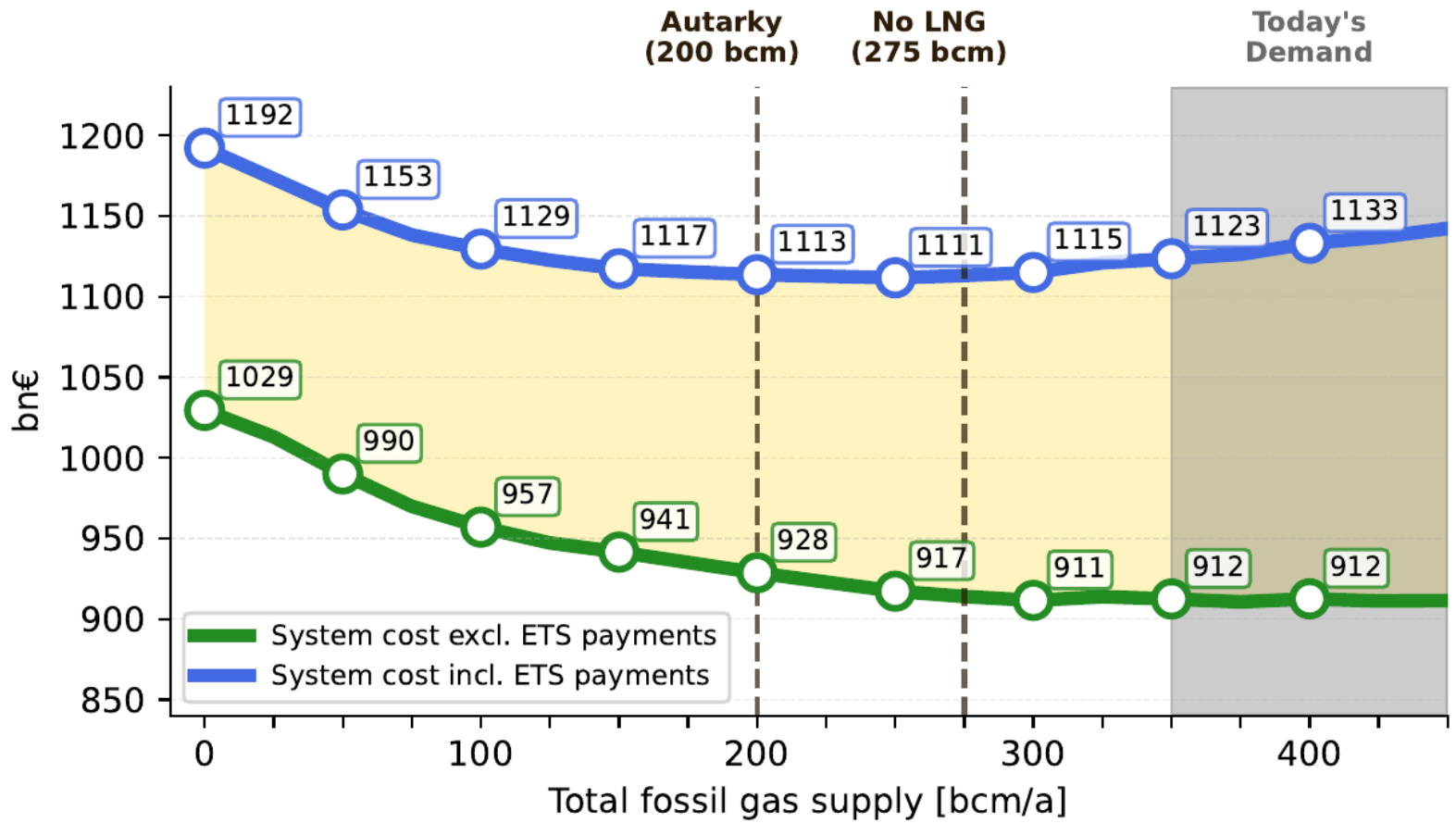
Open high-voltage grid
from OpenStreetMap · Scientific Data 2025

Industrial process heat
endogenous in PyPSA-Eur

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Expected Results: Pricing Energy Security

Quantifying the trade-off between energy security and energy costs

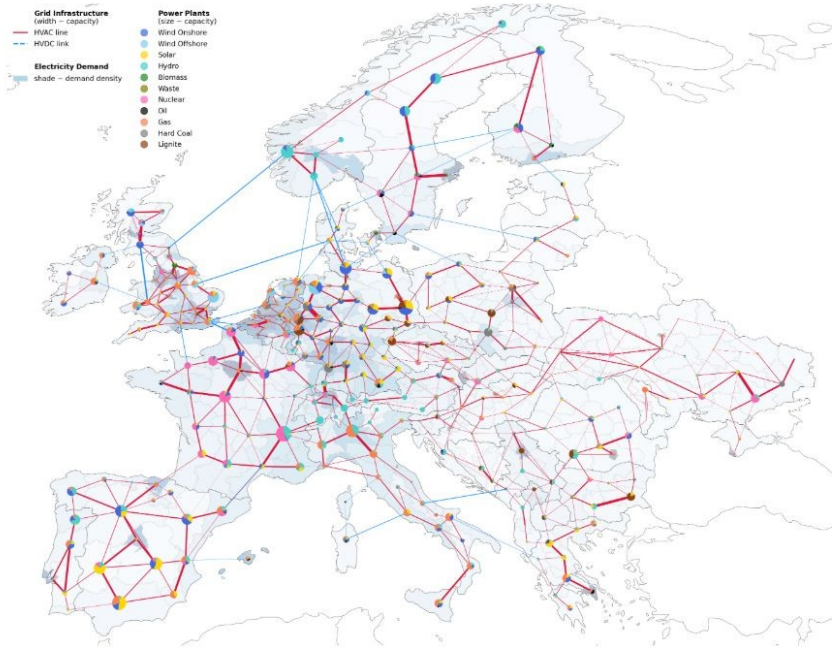


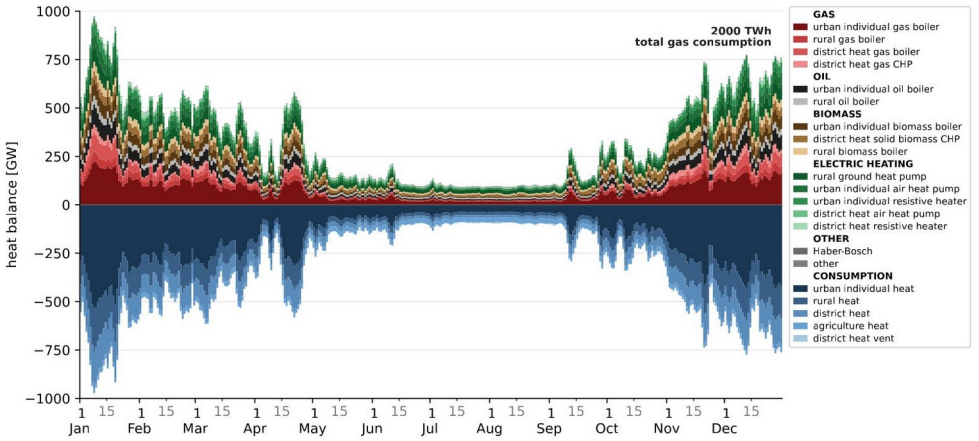
European system cost vs. fossil gas supply · 100 €/tCO_2 · Franken, Riepin & Brown (2026)

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The PyPSA-Eur Model

50 nodes **3-hourly** power · heat · industry · transport · biomass · agriculture





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WORKING PAPER · ARXIV, JULY 2026

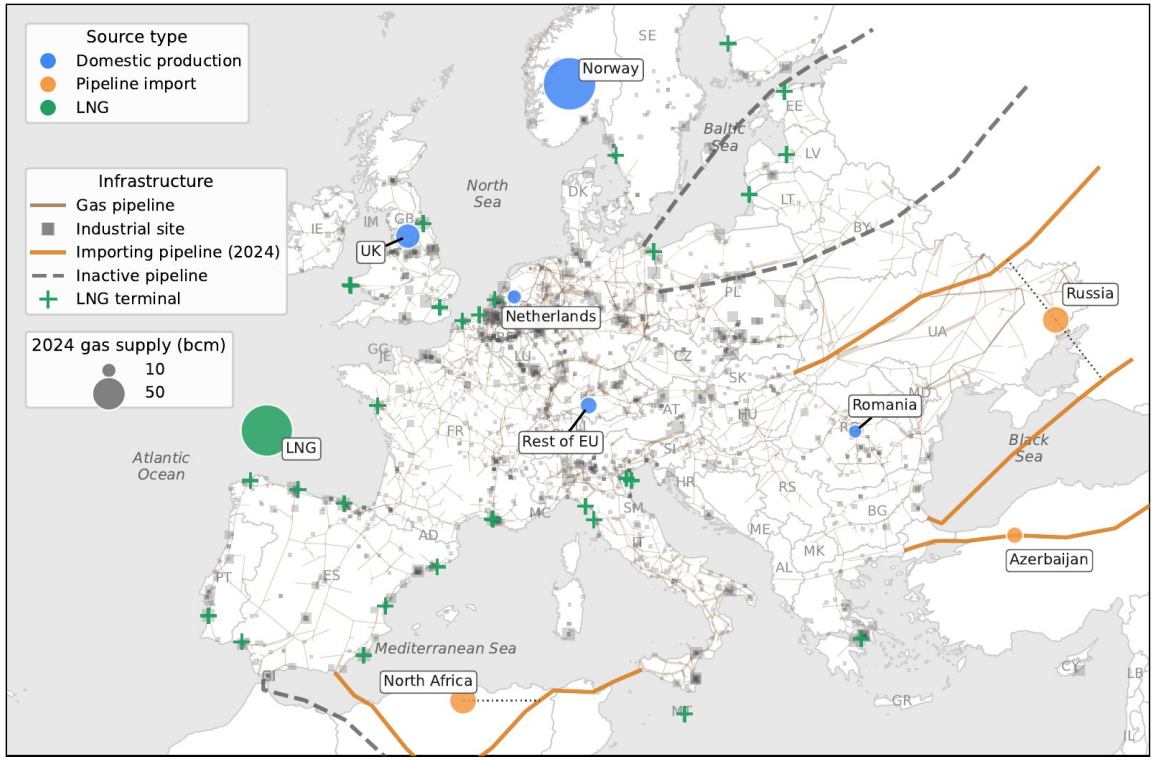
Accelerating fossil gas independence in Europe

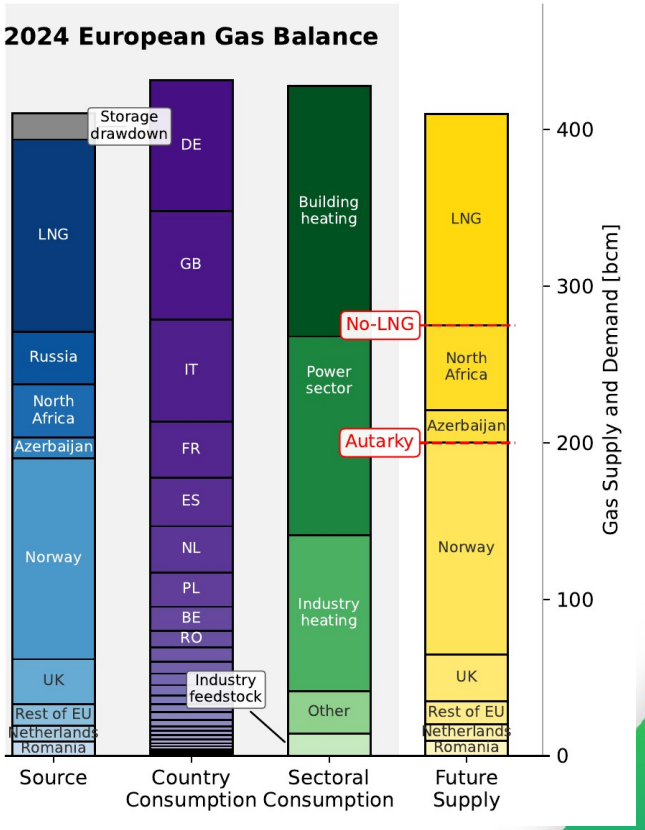
Lukas Franken · Iegor Riepin · Tom Brown

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Europe's Gas Today

Roughly a third each to power, buildings and industry; Europe's own production covers ~200 bcm



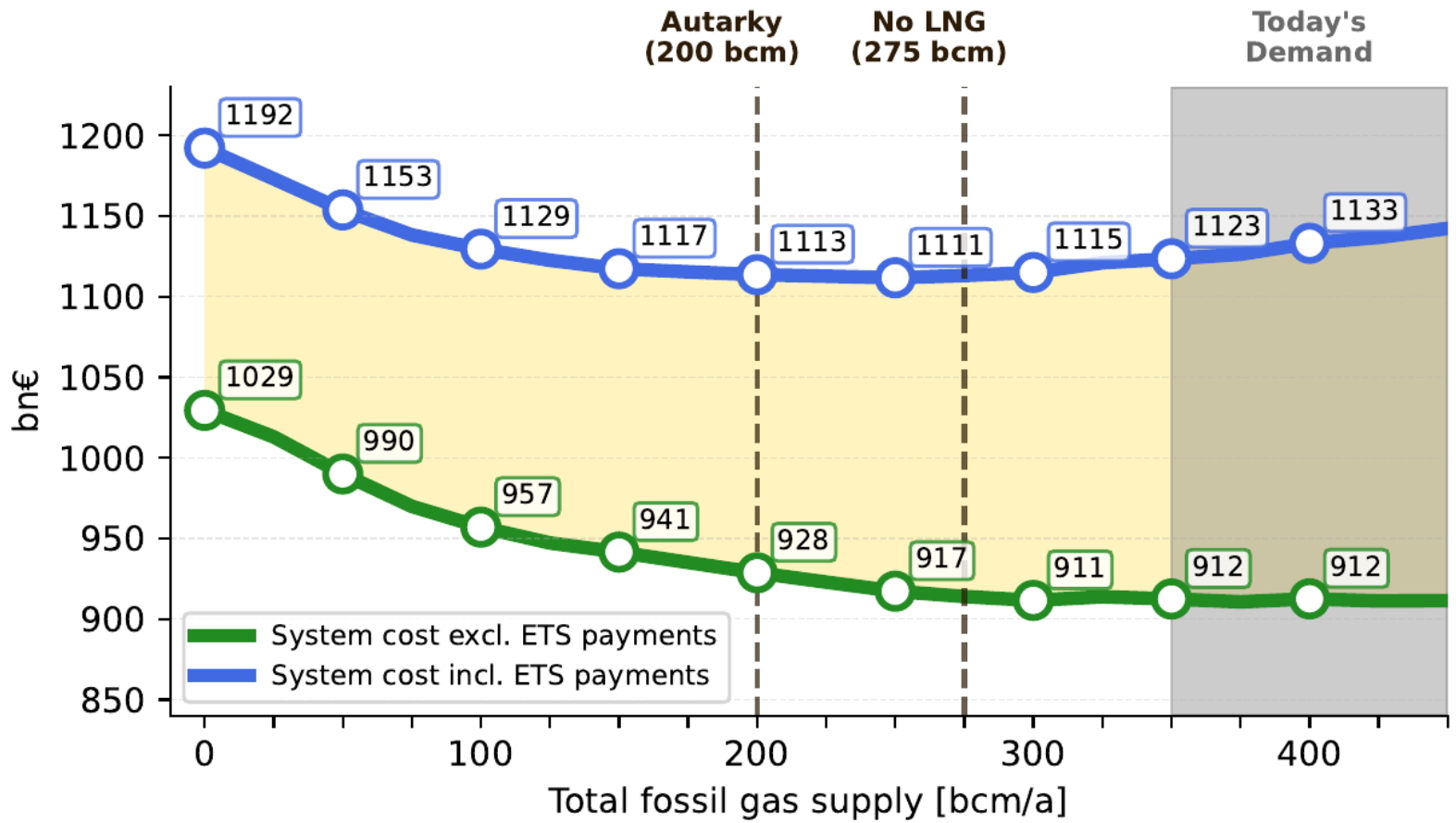


Data: Eurostat (2024), OIES, Hotmaps, OpenStreetMap · Franken, Riepin & Brown (2026)

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The Gas Abatement Curve

No-LNG: +5 bn€/a · Autarky: +16 bn€/a, under 2 % of system cost (excl. ETS payments)

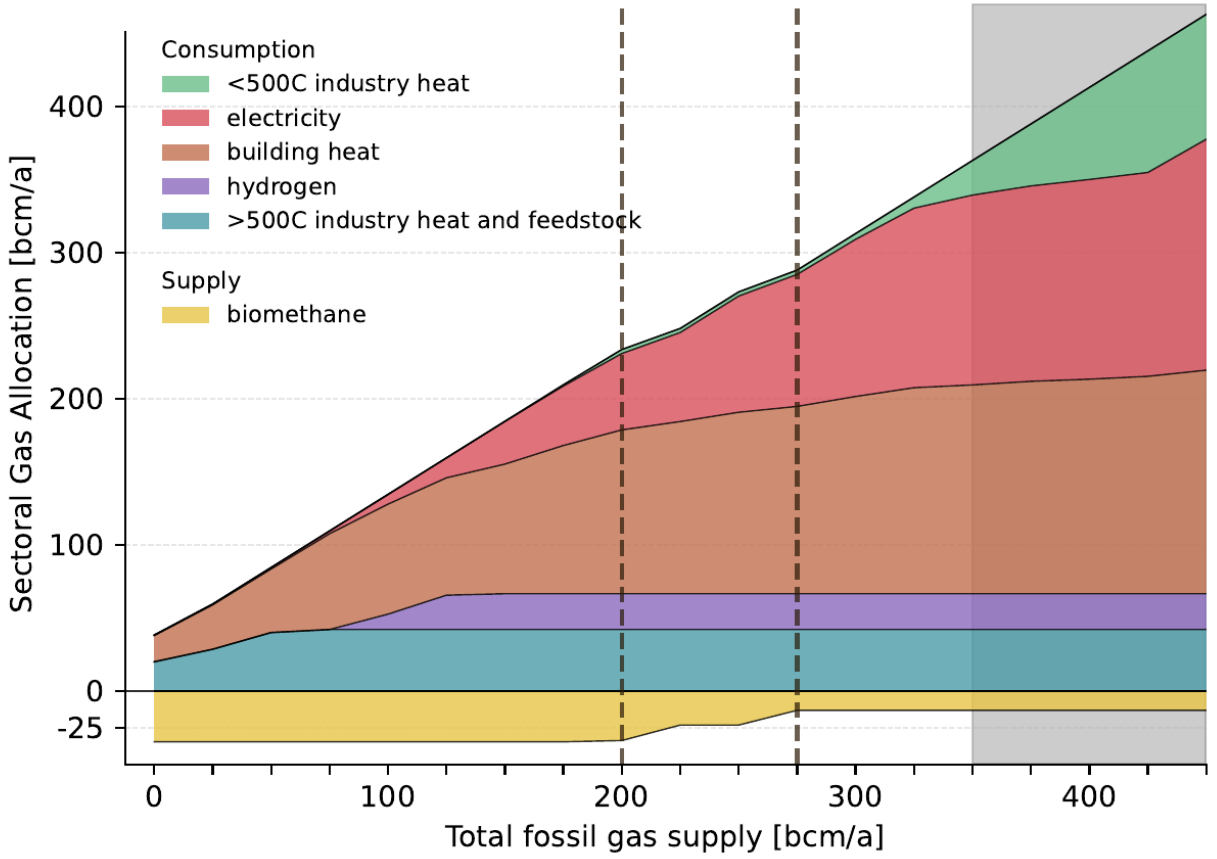


Gas cap 450 → 0 bcm in 25 bcm steps · 100 €/tCO_2 · Franken, Riepin & Brown (2026)

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Where the Gas Goes

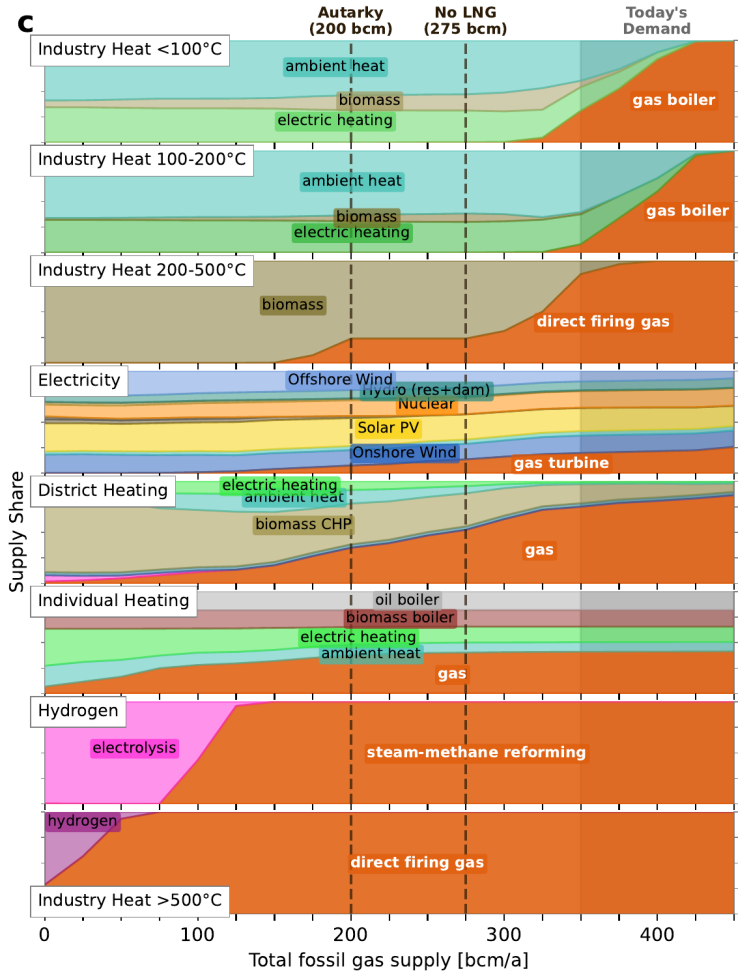
Low-temperature industry heat and power exit first; building heat holds on longest



Dashed lines: Autarky (200 bcm), No-LNG (275 bcm) · grey: today's demand · Franken, Riepin & Brown (2026)

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What Replaces Gas, Sector by Sector



Industry heat < 500 °C

heat pumps, electric heating, biomass

Power

wind, solar and batteries; gas leaves bulk generation at ~250 bcm

Building heat

gradual, with regional hold-outs

Industry > 500 °C

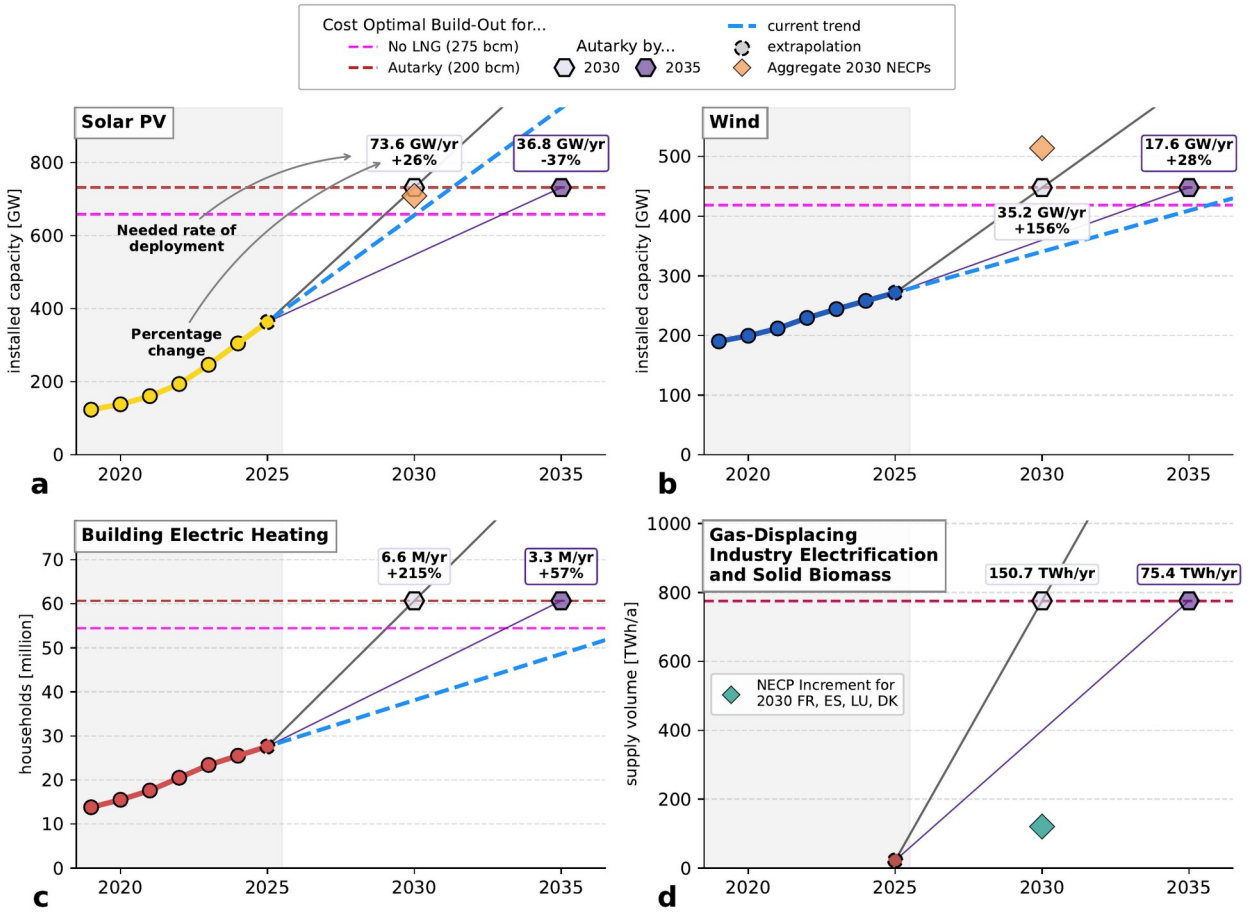
hydrogen, only after > 30 bcm of biomethane

Franken, Riepin & Brown (2026)

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Are We on Track for Autarky?

Autarky needs twice the 2025 wind, solar and heat-pump fleets; power gets there first

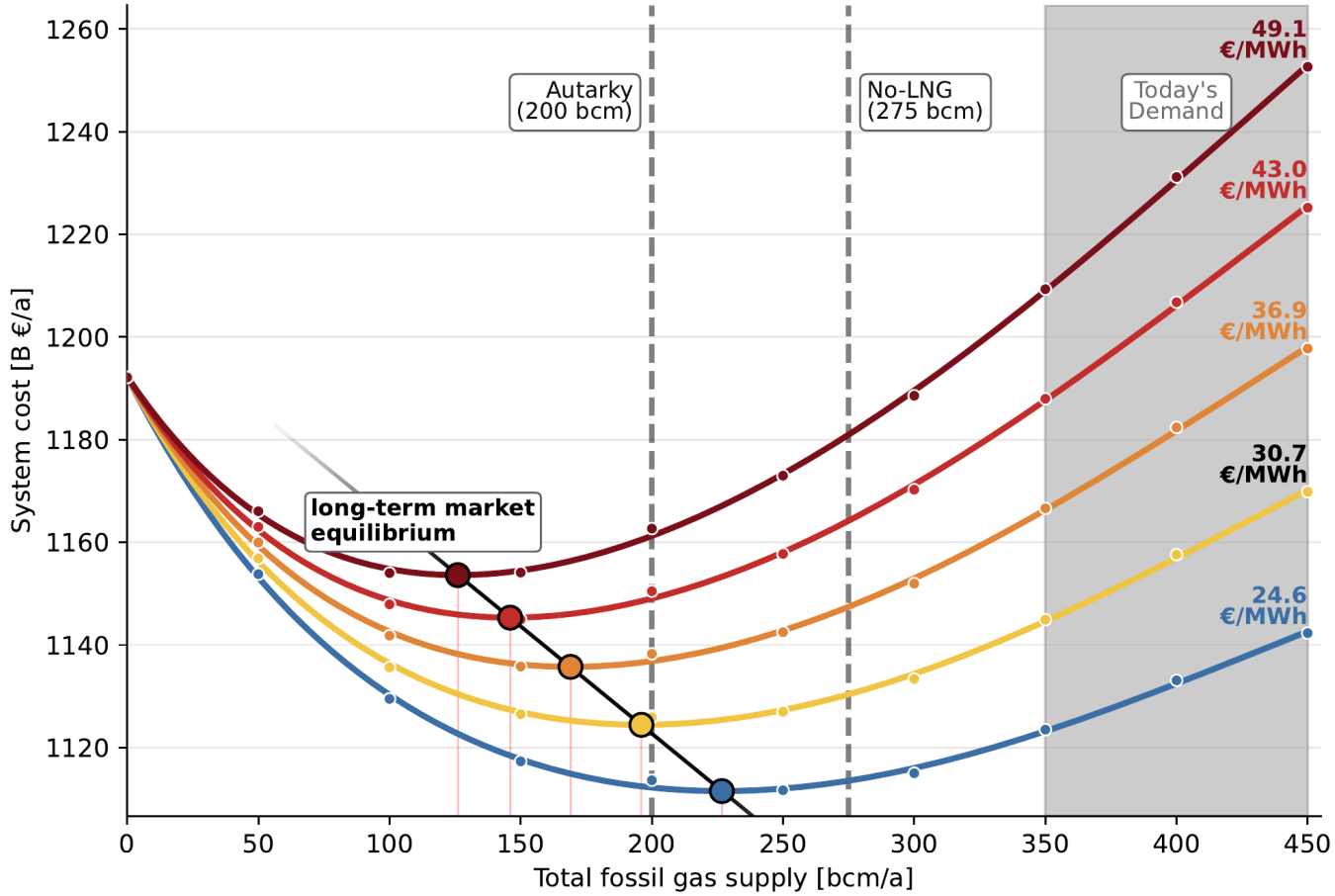


NECP targets 2030: Ember (2024), DESNZ (2024) · Franken, Riepin & Brown (2026)

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Where Gas Demand Settles

At gas prices of 30 €/MWh or more and 100 €/tCO$_2$, autarky is cost-optimal

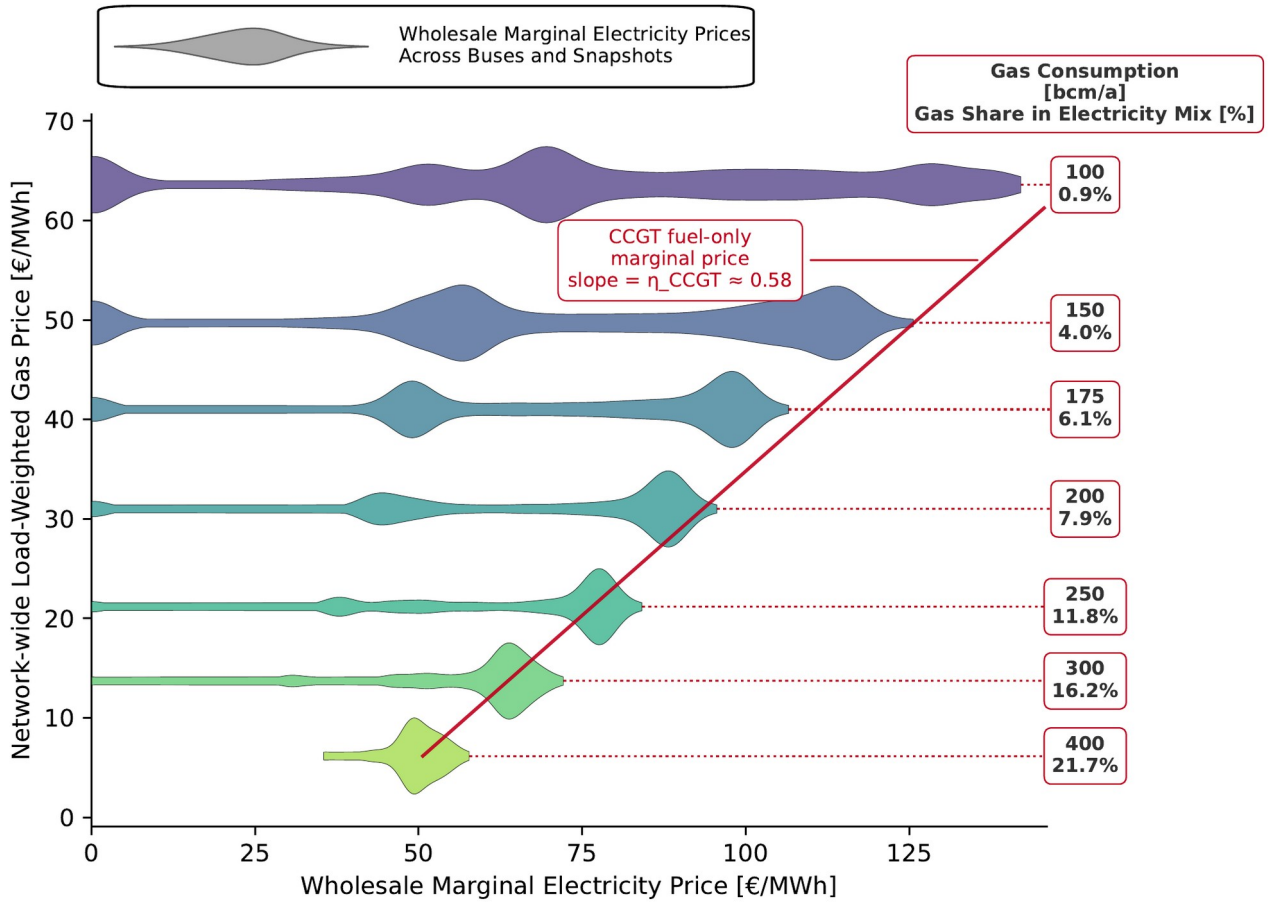


Long-run price elasticity of gas demand ≈ −1.33 · Franken, Riepin & Brown (2026)

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Gas Still Sets the Electricity Price

Even below 8 % of the power mix, gas anchors the price about half the time

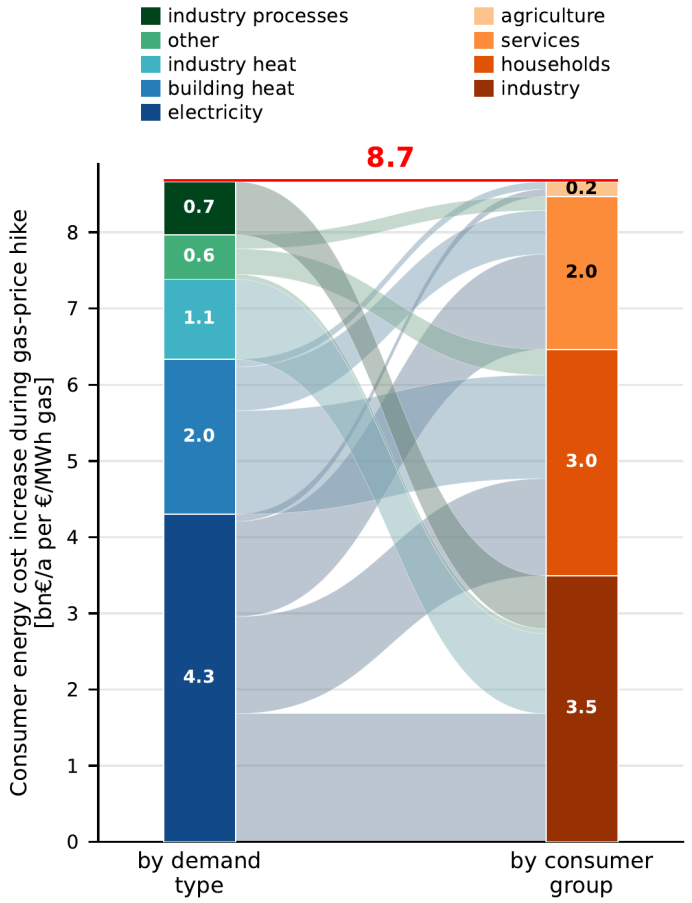


Red line: CCGT fuel-only marginal cost, slope = efficiency 0.58 · Franken, Riepin & Brown (2026)

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the European Union

Gas Price Shocks Reach Every Consumer

+1 €/MWh on gas costs consumers +8.7 bn€ a year, about twice volume × price

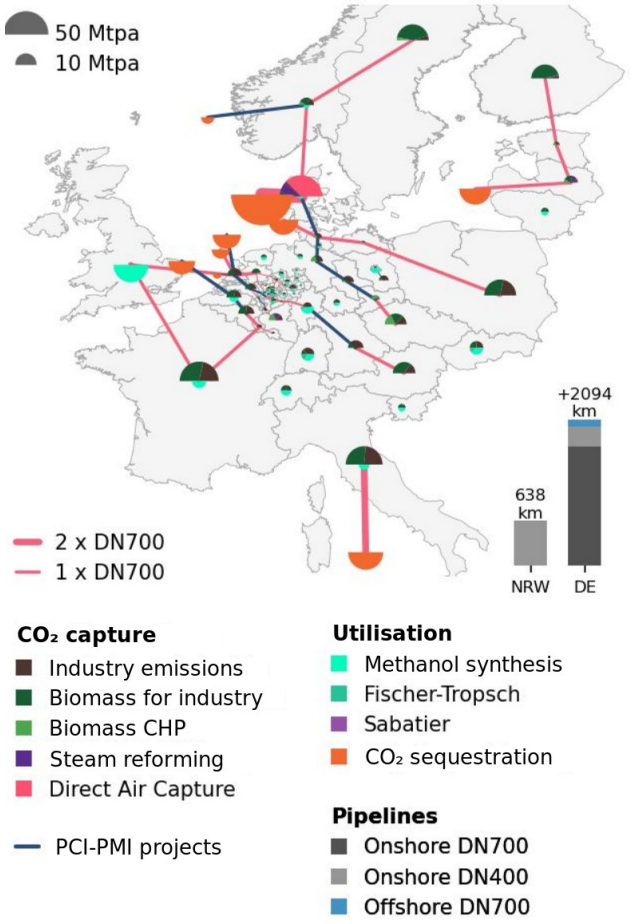


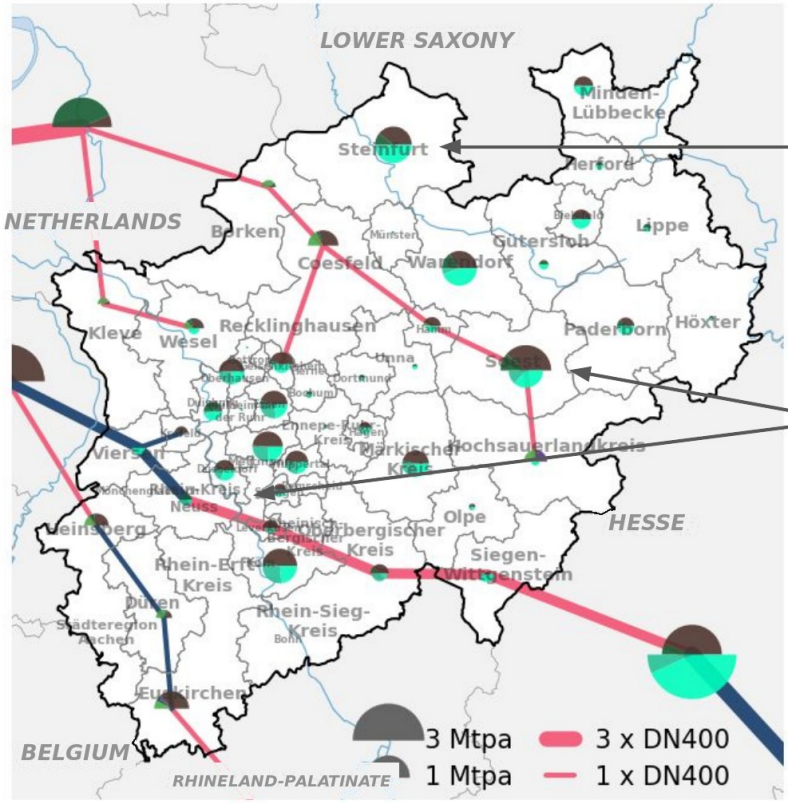
System at 400 bcm/a, operation-only re-optimisation · Franken, Riepin & Brown (2026)

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Other RESILIENT Work: NRW CO_2 Network in 2045

First step for NRW: one or two DN400 feeder pipelines, ~8.9 Mtpa each

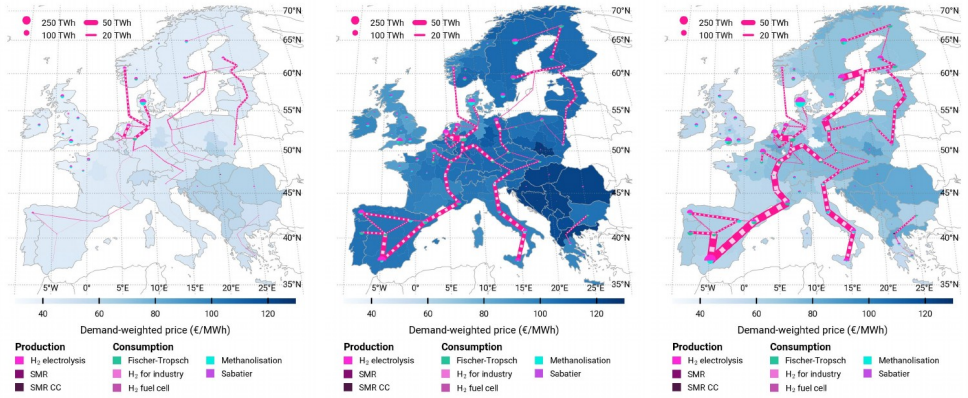




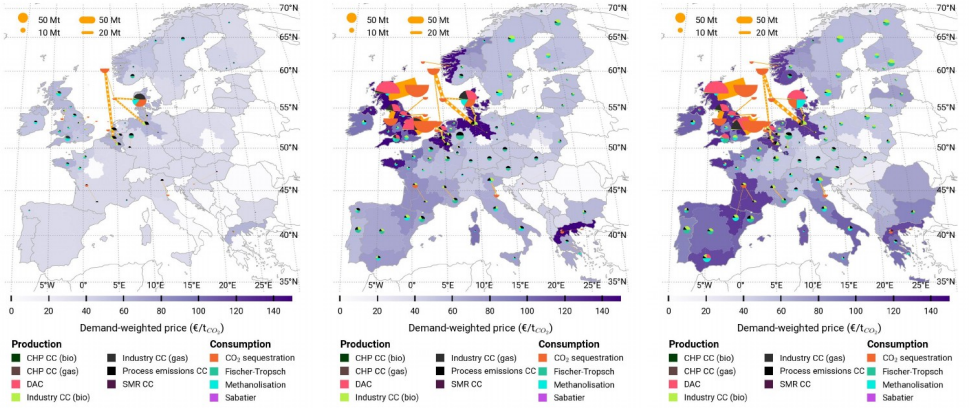
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Other RESILIENT Work: PCI-PMI H_2 & CO_2 Backbone

From 2040, PCI-PMI pipelines cut system costs by more than 26 bn€ a year



(a) H_2 2030. (b) H_2 2040. (c) H_2 2050.



(d) CO_2 2030. (e) CO_2 2040. (f) CO_2 2050.

PCI-PMI long-term scenario · Xiong, Brown & Riepin (2026), Environmental Research Letters

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Other RESILIENT Work: Risk-Averse Planning with CVaR

Stochastic optimisation and CVaR in PyPSA v1.0, the optimisation core under PyPSA-Eur

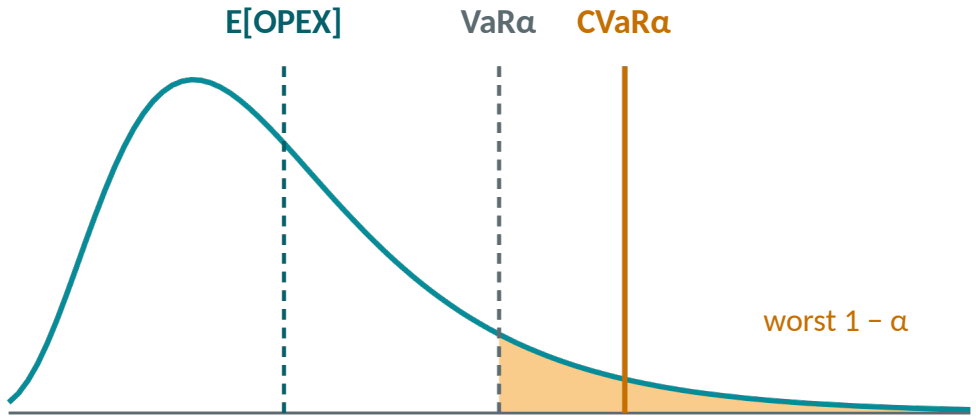
Investment
one plan, shared by all scenarios

Scenario 1
$p_1 \cdot OPEX_1$

Scenario 2
$p_2 \cdot OPEX_2$

Scenario N
p_N · OPEX_N

e.g. weather years, fuel prices, H_2 availability



operating cost across scenarios

$$\min \text{CAPEX} + (1 - \omega) \cdot \text{E[OPEX]} + \omega \cdot \text{CVaR}\alpha\text{[OPEX]}$$

```
n.set_risk_preference(alpha=0.95, omega=0.1)
```

PyPSA pull requests #1154 (stochastic optimisation) and #1345 (CVaR) · RESILIENT

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Thank you!

Contact

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lukas.franken@tu-berlin.de
resilient-project.github.io

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